COMP 4300
Intro to Game Programming

Lecture 1
Course Introduction

Intro to Game Programming

- This course is:
  - Programming that makes video games possible
  - Game Engine / Gameplay Programming
  - C++ / SFML / ImGui
  - Fun!
- This course is not:
  - A game design course
  - A game development course
  - A game engine (Unity, Unreal) tutorial course

C++ Programming

- This course has a lot of programming
- It is slightly more total work than most courses, but there is significant overlap between assignments and project
- Lots of programming, but very rewarding
- Programming has muscle memory, it is very important to write a lot of code

C++ Programming

- Many students have completed the course with no C++ knowledge beforehand
- 4-hour Intro to C++ video to catch up
- Assignment 1 lecture will also have a tutorial on how to set up our development environment for C++ / SFML / ImGui
- Don't be afraid! Modern C++ is awesome

What’s New in 2025?

- This year we have an in-class exam
- SFML3 released, unfortunately not backward compatible with SFML2
- Slight updates to assignments
- Some C++23 quality of life updates

Course Structure

- Lectures:
  - Tuesday / Thursday
  - No mandatory text, lectures are important
- Online Sessions / Help:
  - I’m always available via email
  - I’m available on Discord (Dave’s Courses) (please do not share discord info with public)

The evaluation structure of the course is as follows:

• Assignments		30% (≤ 2 Per Group)
◦ Intro to C++ / SFML / ImGui	(Programming)	
◦ First Game with ECS	(Programming)	
◦ Game: Mega Mario	(Programming)	
◦ Game: Not Zelda	(Programming)	
• Exam	(In-Class, Written)	20% (Solo)
• Final Game Project	(Programming)	50% (≤ 4 Per Group)

Note: <u>You must pass the final project to pass the course.</u>

The work in this course is designed to incrementally build toward the final project. If your grade on the final project is less than 50%, then your overall course grade will be equal to the mark that you received on the final project. If your final project grade is greater than or equal to 50%, your course grade is determined by the scheme above.

Assignment Notes

- Each groups gets one “no questions asked” 48-hour extension (assignments only, use it wisely)
  - TA will notice late assignment and ask if you want to use it
  - No need to contact me beforehand
- Every assignment is due at 11:59pm on the due date
- Unlimited submissions allowed, submit early
- Only last submission is kept
- You lose 5% per hour that the assignment is late, rounded up (1 min = 5%, 62 min = 10%, etc)
- If special circumstance (sick, etc) contact me

Cheating / Plagiarism

- All work is going to be group work
- Keep discussion of assignments and the project entirely within your own group
- We will be checking very closely for cheating, and punishing it strictly
- There are valid excuses to be late, miss work, miss class, etc: but never to cheat

I take academic misconduct very seriously. Anyone found cheating in this course will receive the harshest possible academic penalties. Academic misconduct for this course includes (but is not limited to) the following:

- Handing in any material for evaluation that was done outside you /your group
- Obtaining solutions from ANY non-class source, such anyone outside of your group, previous course offerings, stack overflow, LLMs (unless given specific permission)
- Sharing assignment or exam questions outside of the course for any reason, including assignment sharing websites or online repos such as GitHub
- Reverse engineering any obfuscated solution code that may be given to you

Anti Cheating Measures

- All assignment files will be personalized and sent directly to your email
  - Name, Student Number, email, etc
  - Other steganographic information inserted
- Any files leaked to other groups or online will be traceable back to original students
- Large repository of past assignments to do code difference checking with

Anti Cheating Measures

- We periodically search online / GitHub for code related to the course
  - Have had code removed under DMCA request
- Ensure any repository is made private
- You may not release any source code written for this course at any time for any reason (now or in the future)


OLD MAN YELLS AT CLOUD

AI Generated Code

- It is impossible to give an assignment that undergrads can solve but AI can't
- This presents a very difficult situation for education as these tools improve
- The temptation to press 3 buttons and have your assignment completed for you is very strong, but a dangerous one

AI Generated Code

- The point of university is misunderstood
- The point of this class is not to just collect working assignments; I have lots of those
- The point of this class is to give students the knowledge and skills to solve those assignments on their own
- This is a very important distinction

AI Generated Code

- There are two ‘solutions’ to this problem

1. Catch students who hand in AI generated code and punish them academically
2. Convince students that it is extremely beneficial to them to not use AI tools to automatically solve their problems

Using AI Tools Effectively

- There are two ways to use AI tools

1. Prompt the tools with the assignment problem and get it to solve it for you
2. Solve the problem(s) yourself and use the AI tool to help with small coding issues such as syntax or API calls

Using AI Tools Effectively

- If you use AI tools as an API reference to help you learn how to solve the problems on your own, this is incredibly powerful
    - “How do I sort a vector in C++”
- If you use AI tools to implement the code for you and then just copy/paste to your assignments, this is cheating and very detrimental to the learning process

Using AI Tools Effectively

- If you use AI tools as an API reference to help you learn how to solve the problems on your own, this is incredibly powerful
  - “How do I sort a vector in C++”
- If you use AI tools to implement the code for you and then just copy/paste to your assignments, this is cheating and harmful for learning
- Never under any circumstances submit or push a single line of code from AI you don’t understand

AI Tools in Industry

- You will be using AI tools to implement code for you in a future industry job
- This is to save time by having AI write code that you already understand how to write by yourself
- Even at your jobs, you shouldn't use AI to implement things you don't understand

AI Tools in Industry

- Getting a job requires passing an interview
- You will not have AI in an interview
- If you take shortcuts now while you have this chance to learn, you will regret it
- I speak to many people who comment on how interview skills have gone way down in the past 2-3 years due to AI usage

Final Thoughts on AI

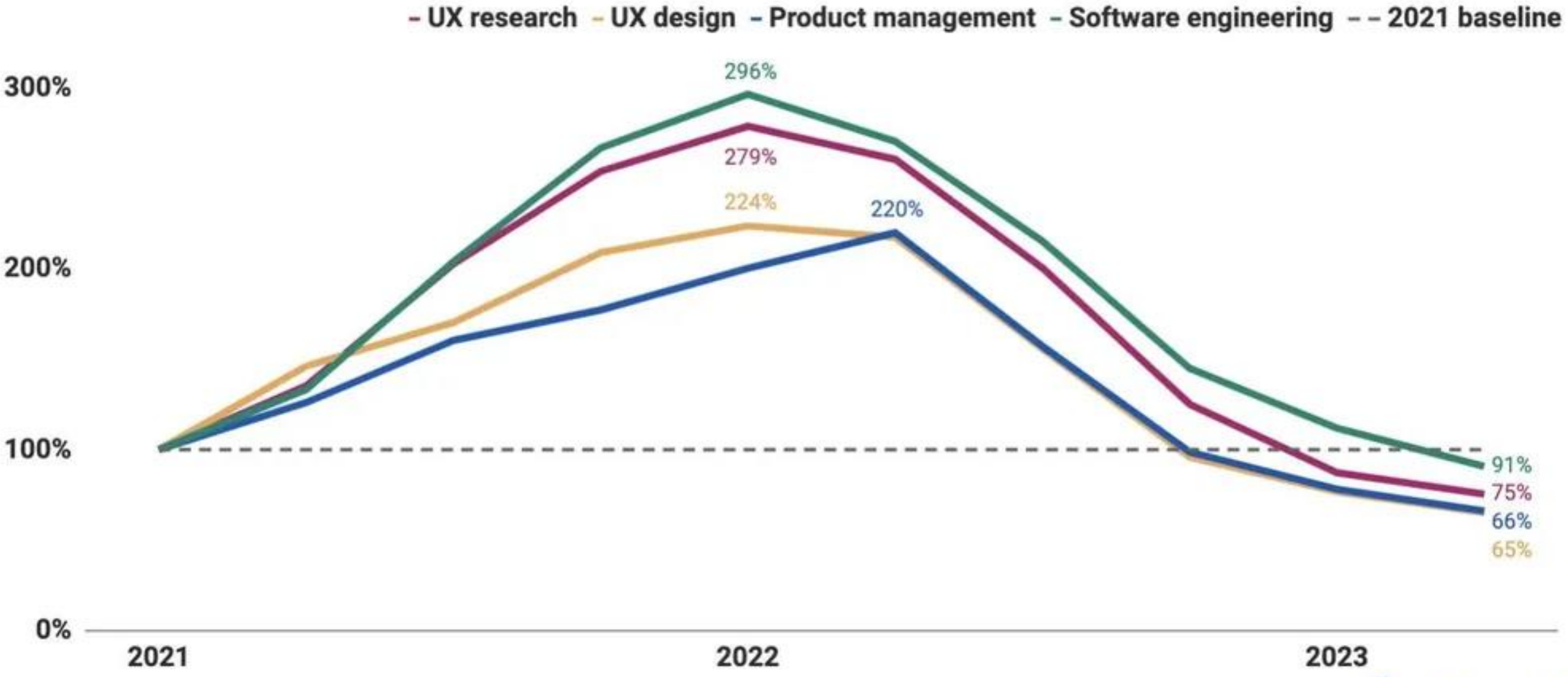
- AI is a useful tool to enhance productivity
- AI can be used to help learn topics
- You will need the knowledge on your own to pass an interview to get a job
- Use school to actually acquire knowledge, practice, and get better at the skills
- Use AI as much as you want at the job

Real-World Consequences

- 2 years ago, I had companies calling me asking for recommendations of good students to hire
- In 2025, my best students are struggling to get internships, co-ops, interviews, etc
- Tech job market in St. John's has gone down significantly in past 1-2 years
- Use this course as a way to practice the skills that you will need to get a job. A piece of paper saying you graduated is no longer sufficient

Change in tech job postings in the US

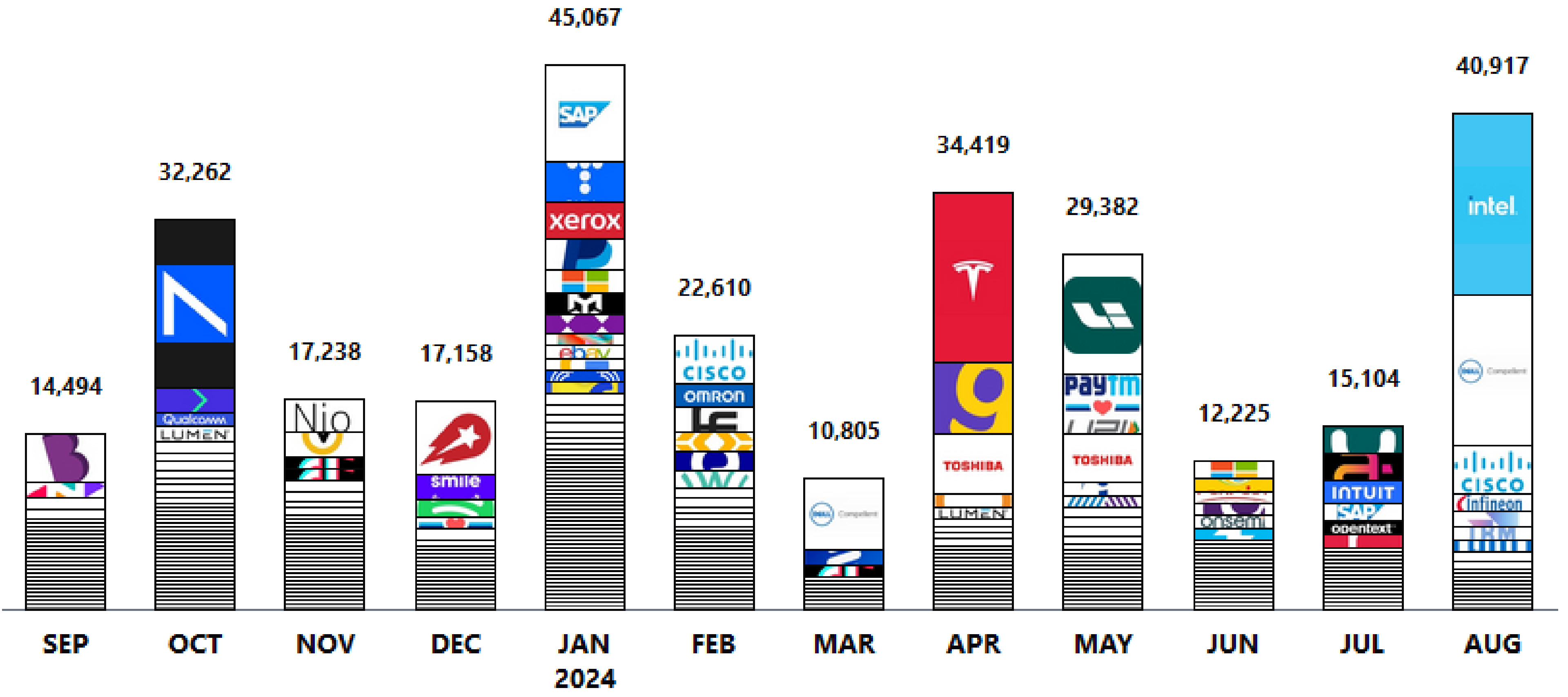
Job postings on Indeed, indexed to the first quarter of 2021



indeed

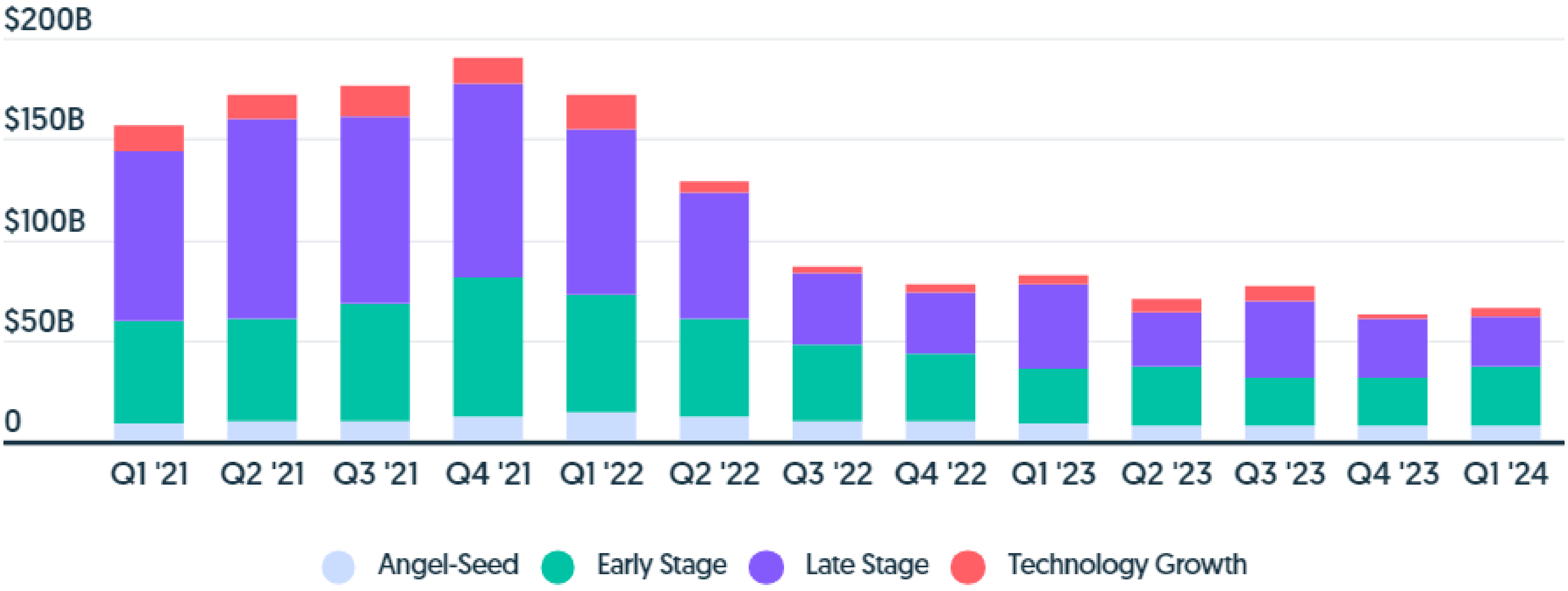
of Tech Employees Let Go

as of August 29, 2024



source: trueup.io/layoffs

Global Venture Dollar Volume Through Q1 2024



QoQ Change (%) YoY Change (%)

A ton of job postings might actually be fake

A new survey found that almost 40% of companies posted a fake job listing this year — and 85% of those companies interviewed candidates for fake jobs

By **Ben Kesslen** Published June 24, 2024

Ghost jobs: What the rise in fake job listings says about the current job market

PUBLISHED THU, AUG 22 2024•6:02 AM EDT

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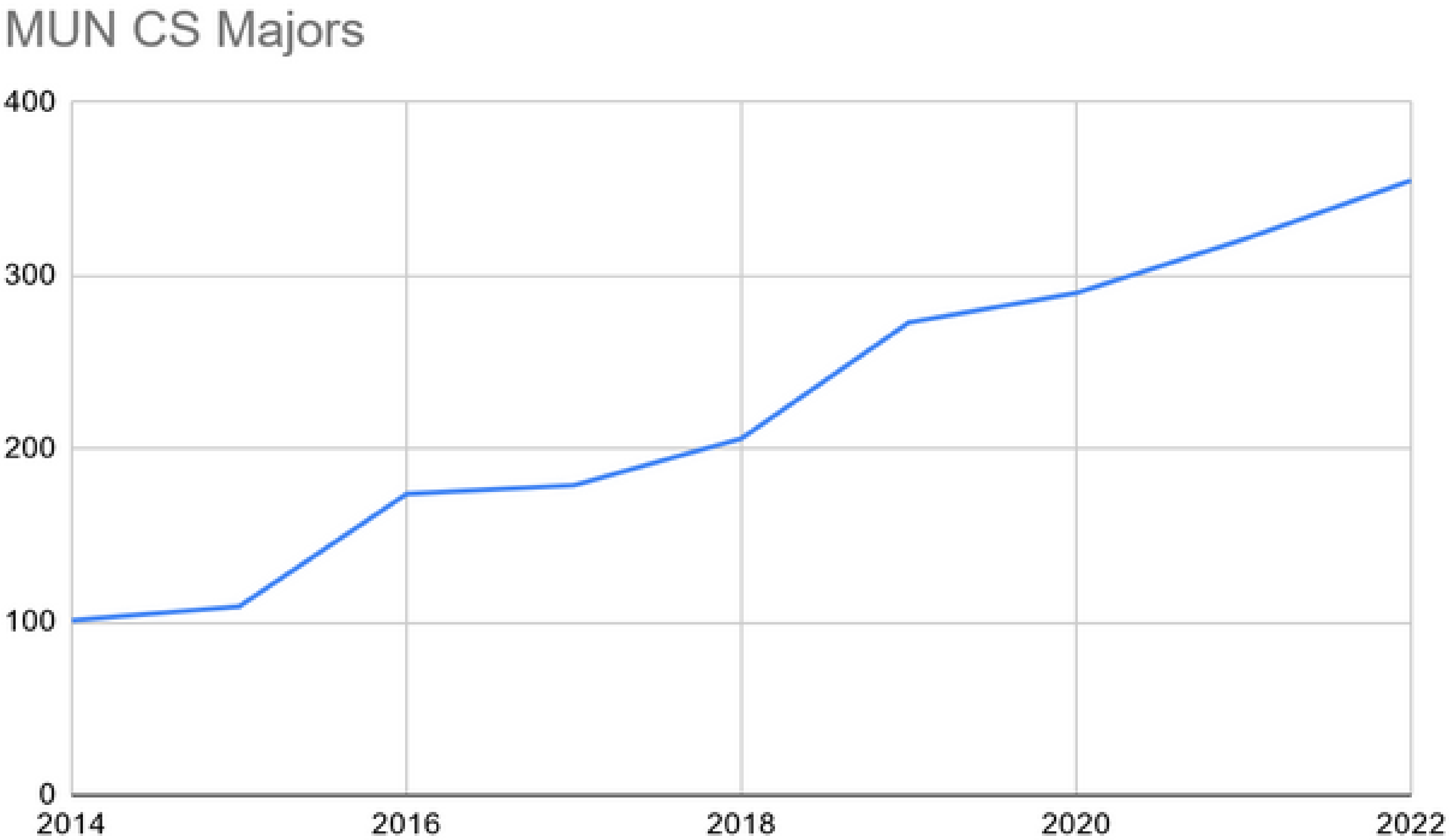
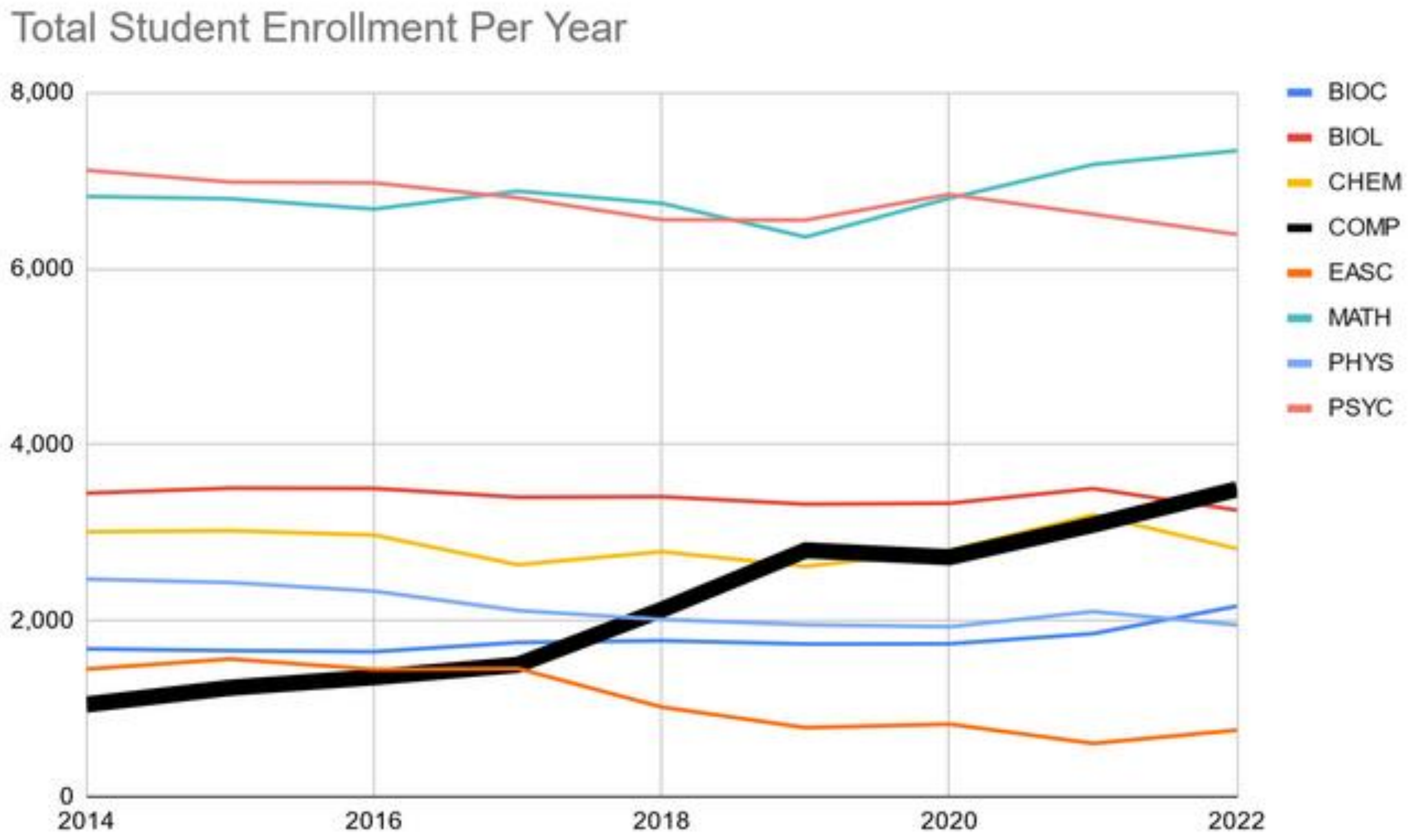
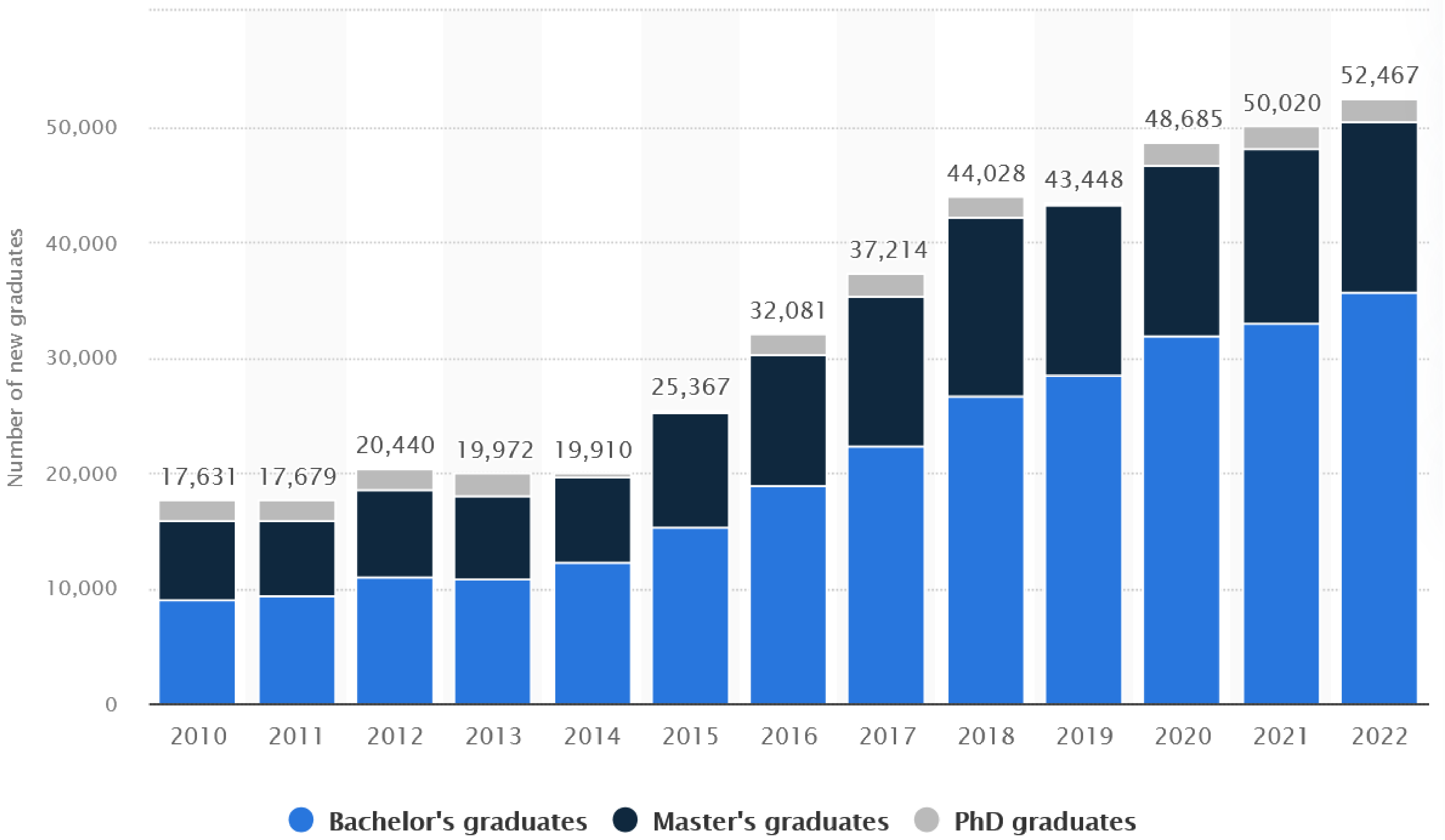


MUN CS Majors
400
300
200
100
0
2014
2016
2018
2020
2022

3) MUN CS enrollment was the lowest of all 8 science departments in 2014, but is now 3rd highest behind only PSYC and MATH (Black Line)





Number of new graduates
50,000
40,000
30,000
20,000
10,000
0
17,631
17,679
20,440
19,972
19,910
25,367
32,081
37,214
44,028
43,448
48,685
50,020
52,467
2010
2011
2012
2013
2014
2015
2016
2017
2018
2019
2020
2021
2022
Bachelor's graduates
Master's graduates
PhD graduates

Local Anecdote

- Local tech company posted a job in June 2024 for a junior software developer
- Got over 300 applications in 48 hours

- When you are in a tech interview, they will know within minutes whether you have the knowledge you claim to have

Portfolio Building

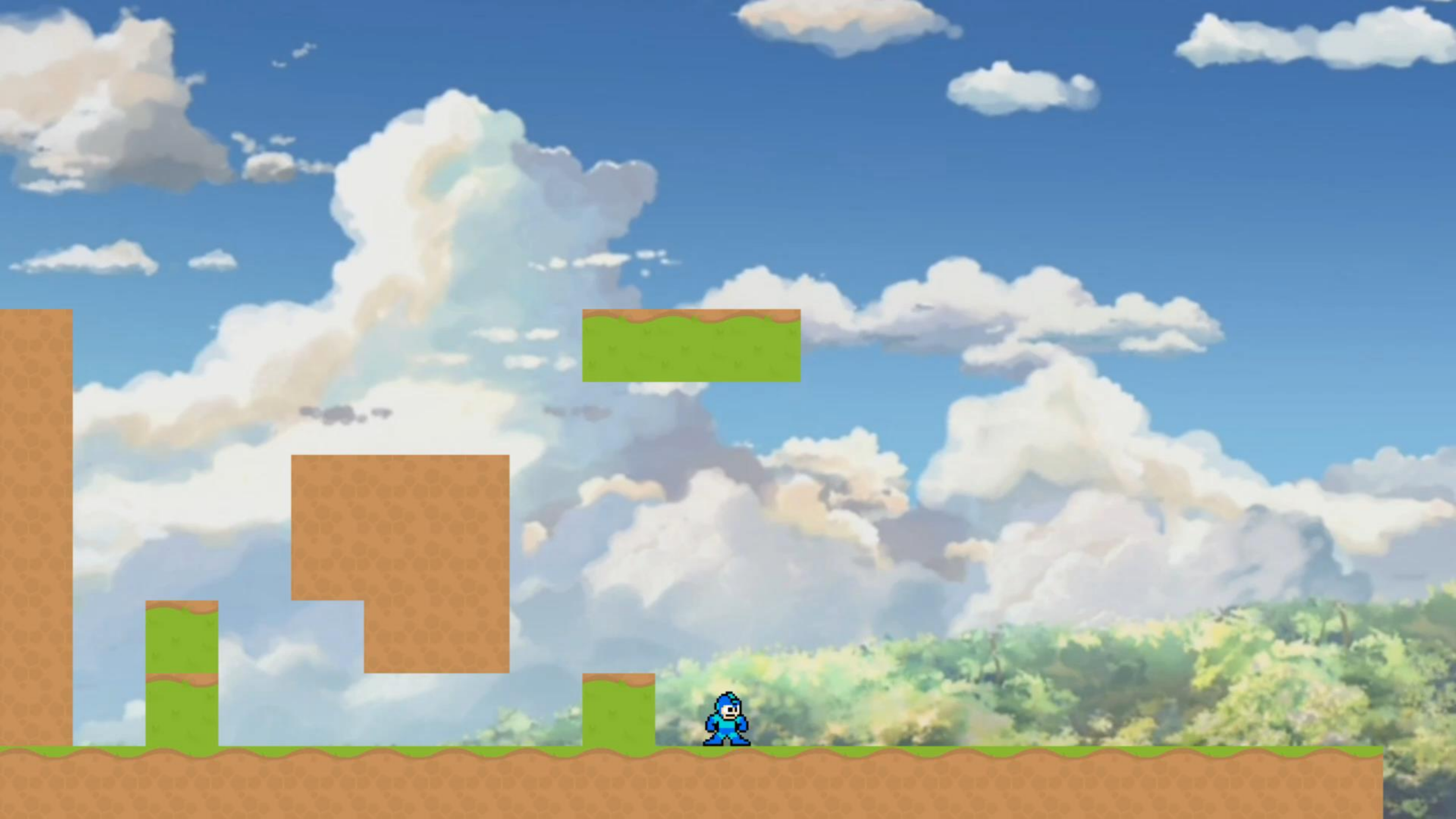
- Even though you cannot release source code online, you can show off your work
- Final project will require a video presentation and game trailer video
- Feel free to share screen shots, videos, or compiled version of your game anywhere
- Course designed to build skills / portfolio

Course Goals?

Course Goal is to Learn:

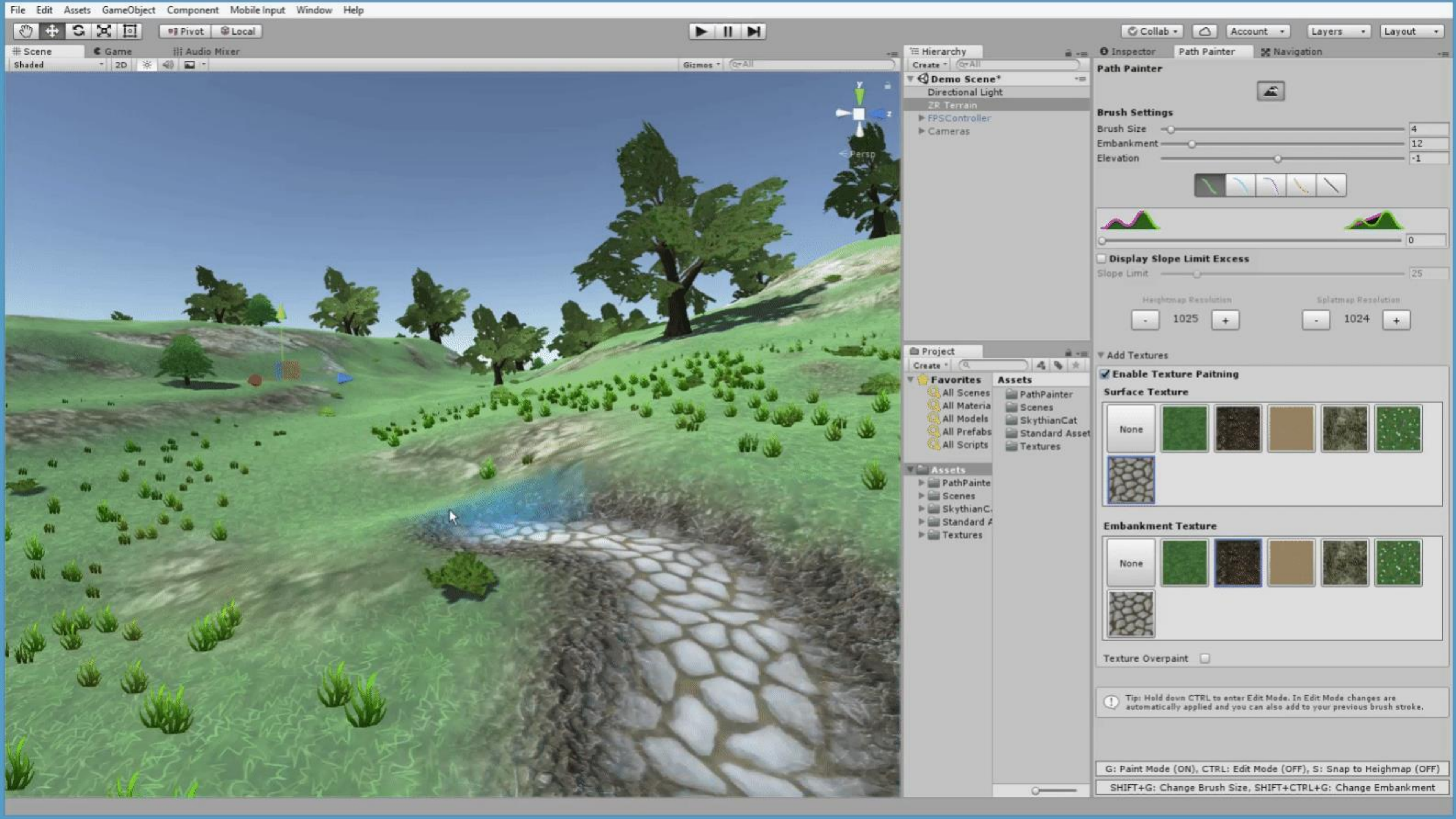
- Game programming from scratch
- Learn the ECS programming architecture
- Game programming patterns
- Software design principles

- Implement several different games

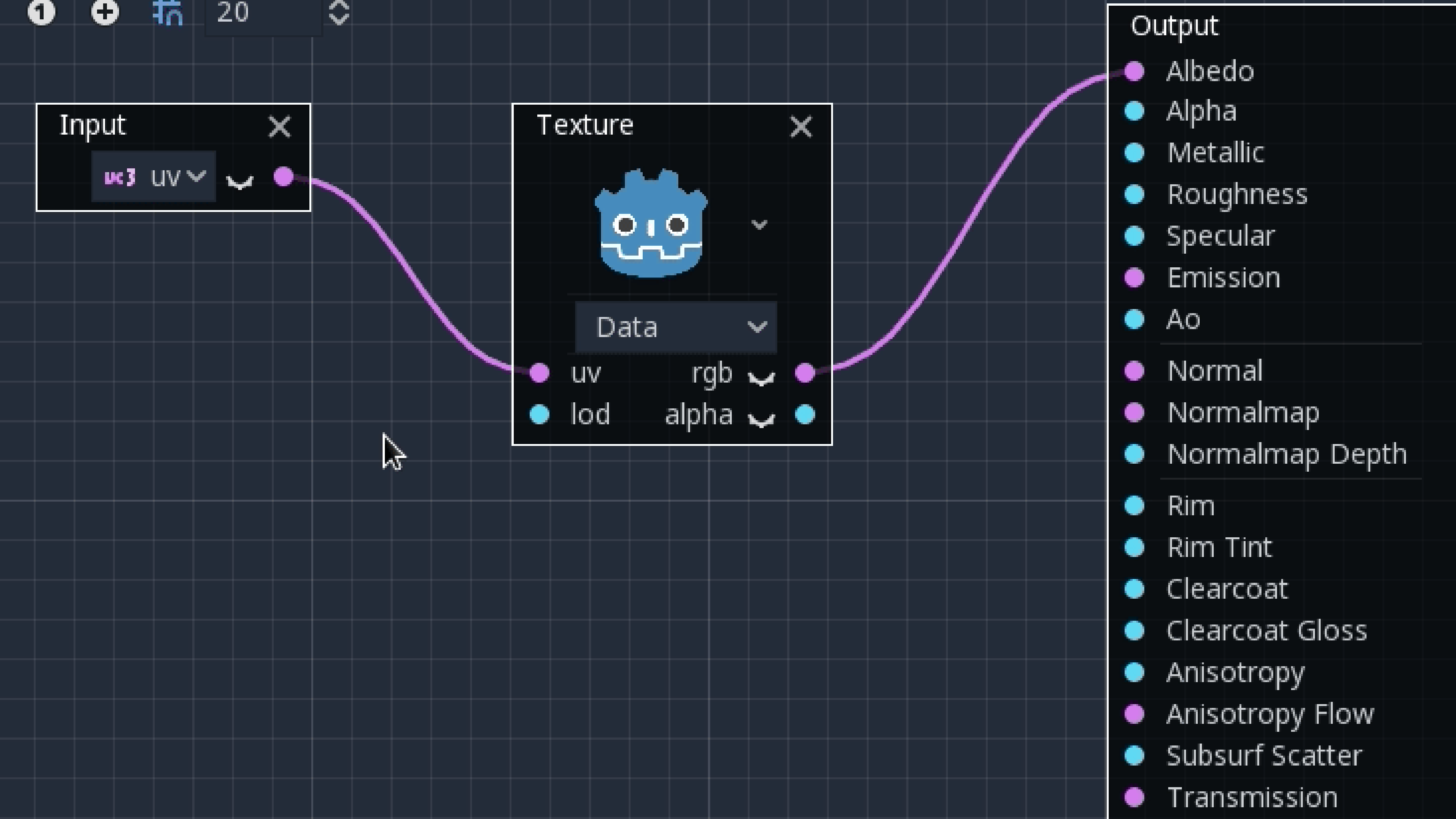


Game Engines

- Many modern game engines exist
- Unity, Unreal, Godot, etc
- Typically have a user interface to interact with, along with coding interface as well


File Edit Assets GameObject Component Mobile Input Window Help
Pivot Local
Collab Account Layers Layout
Scene Game Audio Mixer
Shaded 2D Gizmos
Persp
Hierarchy
Create
Demo Scene*
Directional Light
ZR Terrain
FPSController
Cameras
Project
Create
Favorites
All Scenes
All Materia
All Models
All Prefabs
All Scripts
Assets
PathPainter
Scenes
SkythianCat
Standard Asset
Textures
Inspector Path Painter Navigation
Path Painter
Brush Settings
Brush Size 4
Embankment 12
Elevation -1
0
Display Slope Limit Excess
Slope Limit 25
Heightmap Resolution
1025
Splatmap Resolution
1024
Add Textures
Enable Texture Paitning
Surface Texture
None
Embankment Texture
None
Texture Overpaint
Tip: Hold down CTRL to enter Edit Mode. In Edit Mode changes are automatically applied and you can also add to your previous brush stroke.
G: Paint Mode (ON), CTRL: Edit Mode (OFF), S: Snap to Heightmap (OFF)
SHIFT+G: Change Brush Size, SHIFT+CTRL+G: Change Embankment


Persp


20
Input
uv
Texture
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uv
rgb
lod
alpha
Output
Albedo
Alpha
Metallic
Roughness
Specular
Emission
Ao
Normal
Normalmap
Normalmap Depth
Rim
Rim Tint
Clearcoat
Clearcoat Gloss
Anisotropy
Anisotropy Flow
Subsurf Scatter
Transmission

Game Engine

- We will not make a full unity-like game engine with all of its features
- Our game engine will:
  - Have a C++ ECS programming architecture
  - Be able to program many 2D games very easily
  - Be able to incorporate different game ‘scenes’
  - Make our own a level editor for the project
- We *could* extend our game engine to 3D but we would spend half the course on rendering

Why not Unity / Unreal?

- This is a computer science class, we want to get our hands dirty in the details
- After this course, you will understand how existing game engines work and easily be able to use them, if not create your own
- Teaching an existing engine only teaches you syntax, with very little understanding

Why not Unity / Unreal?

- If you get taught how to use a specific game engine and the company you go to doesn’t use it, you have learned nothing
- If you get taught the underlying way that a game engine works, you will be able to learn any specific game engine easily and know even more about how it works

Course Topics

Introduction to C++ / SFML

- Introduction to C++
  - C++ Syntax / Semantics
  - Standard Template Library (SFML)
- Simple and Fast Multimedia Library (SFML)
  - Basics of SFML
  - Windows, Rendering, Input Handling, etc
- ImGui – Immediate Mode GUI

2D Game Engine Programming

- Game Engine Layout / Architecture
- Main Loop Structure / Tick Rate
- Game States & State Machines
- Asset Loading / Memory Management
- Sprites & Animations, Rendering
- Basic Shaders
- User Input Handling & Event Systems
- Data Oriented Design & Config Files
- Window / Menu / Drag & Drop Systems
- World View: Camera Systems / Viewports

ECS Game Engine Architecture

- Entities, Components, Systems (ECS)
- Architecture, Design, Implementation
- ECS Classes, Structure, Memory Mgmt
- Systems for Gameplay Mechanics

Physics / Math for Games

- Vector Math / Class Implementation
- Game Object Kinematics
  - Position, Velocity, Rotation
  - Acceleration, Gravity, Projectiles
- Collision Detection / Resolution
  - Circles, Rectangles
  - Bounding Boxes
- Mass, Momentum, Inertia

Gameplay Programming

- Basic Artificial Intelligence
  - NPC Behavior, Steering, Path-Finding
- Entity Interactions / Dialogues
- Difficulty Settings / Game Config
- Game Events / Triggers
- Items / Inventory / Weapons
- Saving / Loading Games

Plan of Attack

1. Introduction to C++ / SFML
2. Assignment 1: Simple C++ / SFML / ImGui
   - Simple game main loop
   - Drawing a Sprite to the screen
   - Playing a Sound when something happens
   - (Get used to C++ / SFML / ImGui syntax and compiling)
3. Introduce ECS Game Engine Architecture
4. while (!courseOver)
   - Introduce new topic(s)
   - Assignment: make game using the new topics

Course Balance

- This course will maintain a balance of efficiency vs. usability vs. learnability
- The 'easiest' way of implementing something will be taught first, then if a more efficient way exists, taught later
- There are always trade-offs in any method and no 'best way' to do most things